

Applications Areas

UAO Company was founded in 2015 with the aim of designing, manufacturing and providing technical and engineering services in remote sensing and telematics.

Radio systems in different frequency spectrums, satellite and industrial radar subsystems are among the products of the company.



The Weather business area is divided into three business segments: Meteorology, Energy and Transportation. Real-time, uninterrupted and reliable weather data is the cornerstone for an efficient operation by customers including meteorological institutes, airports and airlines, road and railroad authorities, renewable energy customers, maritime market segments, as well as defense forces.

DigiSonde® enables long time sounding operation, allowing various kinds of sounding, such as ozone monitoring, air pollutants (CO₂, CO, NO₂, dust and particles) due to an embedded expansive interface.



Upper Air Observers Ltd

DigiSonde®



Meteorology



Transportation



Energy



+98 21 562 763 33



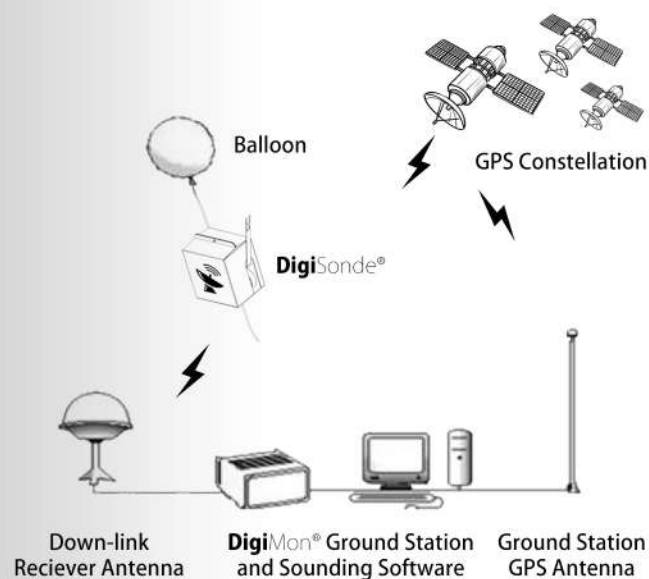
uaobservers.com



IROST Technology Park, Tehran, Iran

General Information

DigiSonde® is an upper air telemetry instrument carried into the atmosphere by a weather balloon that measures various atmospheric parameters and transmits them by radio to a ground station. It can measure or calculate the following variables: pressure, temperature, relative humidity and wind speed and direction from earth surface up to upper air layers.



Specifications

PTU measurement

	Press.	Temp.	RH
Measurement range	1050 ~3 mbar	-90 ~ 60 °C	1 ~99%
Resolution	0.1 mbar	0.1 °C	0.1 %
Accuracy	± 1 mbar	±0.5 °C	± 3%
Response time	<0.1s	<0.5s	<2s
Measurement rate	1 Hz	1 Hz	1 Hz
Factory calibration	Stored in EEPROM		

Wind measurement

Calculation approach	C/A coded GPS
Position resolution	0.1 m
Wind speed resolution	0.1 m/s
Wind direction resolution	0.1 °
Position Accuracy	5 m
Wind speed accuracy	0.2 m/s
Wind direction accuracy	1 °
Measurement rate	1 Hz

Transmitter

Frequency range	400 ~ 406 MHz
Frequency step	10 KHz
Output power	100 mW
Modulation type	Digital FSK 1200 bauds

General

Dimensions	75 x 90 x 110 mm
Weight	200 gr (including batteries)
Battery pack	3 x 1.5 V - 1300 mAh

